About Error and Uncertainty

Everything you need to know about the science of measurements.

"A measurement result is complete only when accompanied by a quantitative statement of its uncertainty. The uncertainty is required in order to decide if the result is adequate for its intended purpose and to ascertain if it is consistent with other similar results." — *National Institute of Standards and Technology*

Introduction

No measuring device can be read to an unlimited number of digits. In addition when we repeat a measurement we often obtain a different value because of changes in conditions that we cannot control. We are therefore uncertain as to the exact values of measurements. These uncertainties make quantities calculated from such measurements uncertain as well.

Finally we will be trying to compare our calculated values with a value from the text in order to verify that the physical principles we are studying are correct. Such comparisons come down to the question "Is the difference between our value and that in the text consistent with the uncertainty in our measurements?".

Percentage Uncertainty

The ratio between a measurement's uncertainty and the measured value is known as the percentage uncertainty.

$$\frac{\delta X}{X} \times 100\% = \text{percentage uncertainty}$$

Propagation of Uncertainties

Let's say we need to make multiple measurements of a system, and then combine those measurements in some algebraic way in our analysis. Depending on how the quantities are combined, we'll need to follow some guidelines known as error propagation.

Addition of two quantities

If, for example, we needed to know the total length of two objects, A and B, measured with different uncertainties, δA and δB, the following equation yields the final measurement and its uncertainty:

$$(A \pm \delta A) + (B \pm \delta B) = (A + B) \pm (\delta A + \delta B)$$

Subtraction of two measurements

Similar to addition:

$$(A \pm \delta A) - (B \pm \delta B) = (A - B) \pm (\delta A + \delta B)$$

Multiplication of two (or more) measurements

Useful for area and volume measurements for example.

$$(A \pm \delta A) \times (B \pm \delta B) = (AB) \left[1 \pm \left(\frac{\delta A}{A} + \frac{\delta B}{B}\right)\right]$$

Division of two (or more) measurements

Useful for density for example.

$$\frac{(A \pm \delta A)}{(B \pm \delta B)} = \left(\frac{A}{B}\right)\left[1 \pm \left(\frac{\delta A}{A} + \frac{\delta B}{B}\right)\right]$$

Inversion

$$\frac{1}{(A \pm \delta A)} = \left(\frac{1}{A}\right)\left[1 \pm \frac{\delta A}{A}\right]$$

Multiplication by a constant

$$k \times (A \pm \delta A) = [kA \pm k\delta A]$$

Square Root

$$\sqrt{(A \pm \delta A)} = \left(\sqrt{A} \pm \frac{\delta A}{2\sqrt{A}}\right)$$

Trig Functions

Trig functions are best dealt with by doing some calculations. For example, let's look at the $\tan^{-1}$ function. Let's say we had to find an angle based on measurements of the opposite and adjacent sides. We would use $\theta = \tan^{-1}\left(\frac{opp}{adj}\right)$. To find the $\delta\theta$, we would need to know the uncertainties in our length measurements.

An example: let's say we measure our opposite side to be $0.5 \pm .01$ m, and our adjacent side to be $2 \pm .02$ m. We would first use the division formula above to obtain the uncertainty in the fraction $\frac{opp}{adj}$:

$$\frac{0.5 \pm .01}{2 \pm .02} = \left(\frac{0.5}{2}\right)\left[1 \pm \left(\frac{.01}{.5} + \frac{.02}{2}\right)\right] = 0.25 \pm 0.0075$$

Then, we would compute the $\tan^{-1}$ of the two values, 0.25 and 0.2575.

$$\tan^{-1}(0.25) = 0.2450 \text{ rad} = 14.04^\circ$$

and

$$\tan^{-1}(0.2575) = 0.2520 \text{ rad} = 14.44^\circ$$

The difference between these is $0.2520 - 0.2450 = 0.0070$. This value is the uncertainty, in radians, of the angular measurement. Converting to degrees gives $0.0070 \text{ rad} \times \frac{180^\circ}{\pi} = 0.401^\circ$. Thus, our measurement of the angle based on measuring the opposite and adjacent sides is: $14 \pm 0.40^\circ$.

Or, in some cases, this can be made even easier by using the small angle approximation: $\sin(\theta) \approx \theta$ if $\theta < .1$ rad. Thus, if we had to find the angle of a triangle with a really long hypotenuse and a really short side:

$$\theta \approx \sin(\theta) = \frac{opp}{hyp}$$

would give us the angle. We could use the division formula from above to calculate the uncertainty in the $\frac{opp}{hyp}$ ratio. This would essentially be the uncertainty in our θ measurement as well (in radians).

Standard Deviation

The definition of the standard deviation of the individual measurements: $x_1, x_2, x_3, \ldots, x_N$ is:

$$\sigma_x = \sqrt{\frac{1}{N}\sum(x_i - \bar{x})^2}$$

The standard deviation of the mean is:

$$\sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{N}}$$